

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE					
Supplementary Winter Examination – 2024					
Course: B.Tech		Branch : Common To All Branches		Semester : I	
Subject Code & Name: BTES105 - Energy and Environment Engineering					
Max Marks: 60		Date:22/02/2025		Duration: 3 Hr.	
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.					
				(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)				12
1	Which of the following is a disadvantage of a steam power station?			CO1	1
	a. High efficiency	b. Low operational cost	c. High fuel consumption		
2	Which renewable energy source is utilized in the Ocean Thermal Energy Conversion (OTEC)?			CO1	1
	a. Solar	b. Wind	c. Tidal		
3	Which of the following is NOT a component of a gas turbine power plant?			CO2	1
	a. Compressor	b. Heat exchanger	c. Reactor		
4	What is the primary environmental concern associated with nuclear power plants?			CO2	1
	a. High greenhouse gas emissions	b. Waste disposal	c. Water pollution		
5	The maximum energy efficiency principle is part of which aspect of energy?			CO2	1
	a. Energy conservation	b. Energy generation	c. Energy transmission		
6	Which method is used to conserve energy in air conditioning systems?			CO3	1

	a. Reducing compressor load	b. Increasing fan speed	c. Using high-efficiency motors	d. Increasing cooling temperature		
7	Which of the following is a source of water pollution?				CO3	1
	a. Smoke from factories	b. Industrial discharge into rivers	c. Light emissions from vehicles	d. Noise from construction sites		
8	Which of the following is NOT a cause of air pollution?				CO3	1
	a. Carbon emissions from vehicles	b. Smoke from forest fires	c. Chlorofluorocarbons (CFCs)	d. Noise from traffic		
9	Biomass energy is primarily derived from:				CO4	1
	a. Fossil fuels	b. Living organisms	c. Wind	d. Water		
10	In energy conservation, what is the main advantage of using energy-efficient lighting systems?				CO4	1
	a. Increased power consumption	b. Higher installation costs	c. Reduced energy consumption	d. Increased brightness		
11	What type of pollution is mainly associated with the burning of fossil fuels?				CO5	1
	a. Water pollution	b. Air pollution	c. Noise pollution	d. Soil pollution		
12	Which energy source involves the conversion of thermal energy from the ocean into electricity?				CO5	1
	a. Wind energy	b. Ocean Thermal Energy Conversion (OTEC)	c. Solar energy	d. Biogas		
Q. 2 Solve the following.						12
A)	Explain the schematic arrangement, advantages, and disadvantages of a steam power station.				CO1	6
B)	Describe the environmental aspects for selecting the sites and locations of power plants.				CO1	6

Q.3	Solve the following.		12
A)	Discuss the schematic arrangement, advantages, and disadvantages of a solar power plant.	CO2	6
B)	Explain the working principle of Magneto Hydro Dynamics (MHD) power generation.	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	Explain the scope for energy conservation in electric furnaces and its benefits.	CO3	6
B)	Describe the methods and techniques of energy conservation in ventilation systems.	CO3	6
C)	Discuss the advantages and disadvantages of biogas as a renewable energy source.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Explain the effects of air pollution on human health and the environment.	CO3	6
B)	Discuss the control measures for particulate emissions in industries.	CO4	6
C)	Describe the control measures for noise pollution in urban areas.	CO5	6
Q. 6	Solve Any Two of the following.		12
A)	Explain the effects and control measures of water pollution caused by industrial discharge.	CO3	6
B)	Discuss the disposal methods of solid wastes and bio-medical wastes.	CO4	6
C)	Explain the concept of thermal pollution and its environmental impact.	CO5	6
	*** End ***		